

Christopher Michael Camplin

Data Visualization Engineer | Frontend Developer | 3D Specialist

GitHub | Shadertoy

Professional summary

Creative and highly skilled Data Visualization Engineer with over a decade of experience in crafting interactive, visually compelling, and performant web applications. Proficient in React, D3.js, Three.js, and WebGL, with a proven track record of developing innovative 3D and data visualization projects for high-profile clients like Siemens, Cisco, Blancpain, and The Economist. Passionate about transforming complex datasets into intuitive visual narratives and pushing the boundaries of modern web technologies.

Technical Expertise

Data Visualization: D3.js, Canvas, WebGL, Three.js, React Three Fiber

Frontend Development: JavaScript (ES6+), TypeScript, React, Redux, Next.js, Framer Motion

3D & Graphics: GLSL Shaders, R3F, Instancing, Shaders, Scrollytelling Animations

Backend Development: Node.js, GraphQL, MongoDB

Other Tools: HTML5, CSS3, Styled-Components, SASS, Git, Webpack, Stable diffusion & ComfyUI

Key Projects

Blancpain - Beyond the Surface (2024)

- Built a statically generated React app using Next.js, R3F, and Framer.
- Developed custom Node.js scripts to process geoJSON data for marine boundaries, optimizing build time.
- Designed a bespoke full-screen fragment shader for homepage transitions and wave animations responsive to scroll velocity.

[Live Demo](#)

Lloyd's Register - Global Maritime Trends 2050 (2023)

- Developed a 3D globe visualization using Next.js and R3F, working closely with 3D designers.
- Implemented instancing techniques to reduce draw calls and enhance performance.

[Live Demo](#)

The Economist - Innovation Quotient (2023)

- Led a team to build a dynamic scrollytelling experience featuring 3D animations and interactive data visualizations.

[Live Demo](#)

Cisco - Bernabéu Reimagined (2023)

- Created a visually engaging 3D scrollytelling page to showcase Cisco's stadium technology using Three.js.

[Live Demo](#)

Future-Ready Business (2022)

- Designed a questionnaire-driven 3D data visualization using D3.js and custom shaders for dynamic chart updates.

[Live Demo](#)

Health Inclusivity Index (2021)

- Engineered a unique 3D visualization representing an iris shape using WebGL and R3F.

[Live Demo](#)

Professional Experience

The Economist Group

Senior Frontend Developer / tech lead

Oct 2020 – Present

- Led the development of interactive data visualizations and 3D experiences for global brands.
- Specialized in building performant, scalable, and visually rich applications using modern web technologies.
- Lead offshore teams and collaborated with dev ops, design and data teams to ensure technical feasibility and alignment with project goals.

Signal Noise

Senior Frontend Developer

Oct 2018 – Sep 2020

- Led the development of interactive data visualizations.
- Specialized in building performant, scalable, and visually rich SPAs using modern web technologies.
- Collaborated with design and data teams to ensure technical feasibility and alignment with project goals.

Hugo & Cat

Frontend Developer

Aug 2018 – Oct 2018

- Developed React apps and modular components for Tourism Ireland and Sony.
- Integrated APIs and implemented dynamic, responsive interfaces using modern JS frameworks.

Tangent

Frontend Developer

Mar 2017 – Jul 2018

- Created custom UI components and map-based features for UK Power Networks and Freesat.
- Focused on accessible, user-friendly, and maintainable codebases.

Independent Consultant

Web Developer

Feb 2012 – Mar 2017

- Provided bespoke web development services to various clients and agencies, focusing on eCommerce and interactive applications.

1000heads

Web Developer

Feb 2010 – Mar 2012

- The role at social media marketing company 1000heads included developing fully responsive js, html 5 & CSS3 web & microsites as well as developing Facebook apps using the Facebook API, PHP & MySQL to connect the tabs, interact with Facebook & collect data for competitions. I also learned to use Django, and honed my Wordpress skills & learned a great deal about social media.

Apple Europe

Web Developer

Feb 2008 – Mar 2010

- I worked for Apple Corp. on the Apple.com website where I was part of a team working on the European equivalents of the main US site, which now includes more than 50 countries. Day to day tasks involved bug fixing, localization of content, Photoshop work, video processing work, deploying sites, building pages and repo management using both SVN & FTP. The scripting languages mostly used were HTML5, CSS 3 and JavaScript libraries (prototype, JQuery & Scriptaculous).

Education

BSc Hons in Communication and Digital Media

Brighton University

A Levels: ICT, Media Studies, Communication Studies

Hobbies & Interests

- Exploring creative coding with GLSL, WebGL, and audio-visual integrations.
- Crafting electronic music and experimenting with Stable Diffusion & generative AI.
- Enthusiastic about cycling, hiking, surfing, and exploring political discourse.